homework1

3. Convert the following numbers to 12-bit signed binary numbers.

a) $135_{10} = 128_{10} + 7_{10} = 1000\ 0000_2 + 0111_2 = 0000\ 1000\ 0111_2$

b) $-135_{10} = 1000\ 1000\ 0111_2$(原码) $= 1111\ 0111\ 1000_2$(反码)
$= 1111\ 0111\ 1001_2$(补码)

c) $35F_{16} = 0011\ 0101\ 1111_2$

d) -576_{10}
$576_{10} = 512_{10} + 64_{10} = 10\ 0000\ 0000_2 + 0100\ 0000_2 = 0010\ 0100\ 0000_2$
$-576_{10} = 1010\ 0100\ 0000_2$(原码) $= 1101\ 1011\ 1111_2$(反码)
$= 1101\ 1100\ 0000_2$(补码)

4.完成下列8位无符号二进制数的计算，并给出相应的十进制数。

a) $1010\ 1010_2 + 0011\ 1011_2 = 1110\ 0101_2$
$= 128+64+32+4+1 = 229$

b) $0011\ 1101_2 + 0111\ 0100_2 = 1011\ 0001_2$
$= 128+32+16+1 = 177$

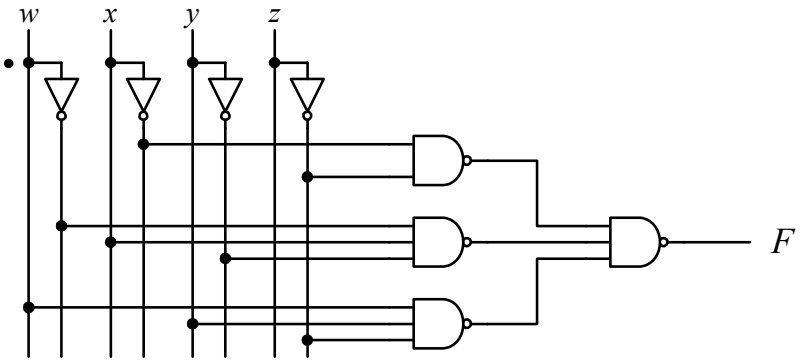
c) $1110\ 0011_2 + 1111\ 0011_2 = 1\ 1101\ 0110_2$
$= 256+128+64+16+4+2 = 470$

d) $1100\ 0111_2 + 1001\ 0110_2 = 1\ 0101\ 1101_2$
$= 256+64+16+8+4+1 = 349$

5. Derive the truth table for the function
F(x,y,z) = [(x+y') + (xz)'] (xy' + y'z)

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

6. Derive the truth table and equation for the following circuit. Do not simplify the equation.



F=((x'z')'(w'xy')'(wyz')')'

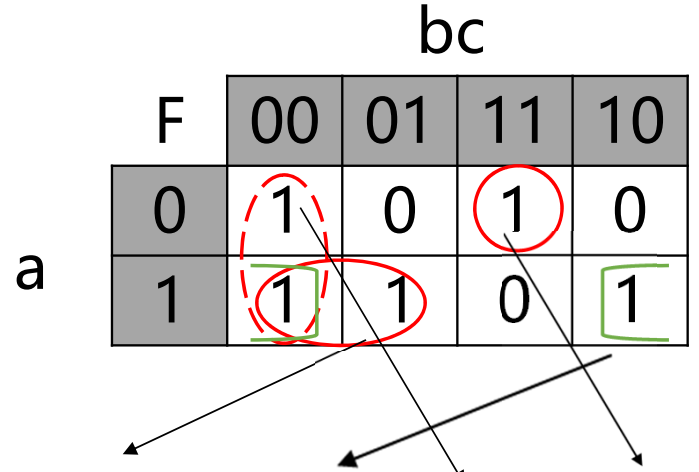
w	x	y	z	F
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0

w	x	y	z	F
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0

7. Derive the Boolean function for the following truth table, and simplify it by using Karnaugh Maps.

a	b	c	F
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

F=a'b'c'+a'bc+ab'c'+ab'c+abc'



F=ab'+ac'+b'c'+a'bc

8. Derive the Boolean function for the following truth table, and simplify it by using Karnaugh Maps.

F=w'x'y'z'+w'x'yz'+w'xy'z'+w'xy'z+wx'y'z'+wx'yz'+wxyz'

w	x	y	z	F
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0

w	x	y	z	F
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0

